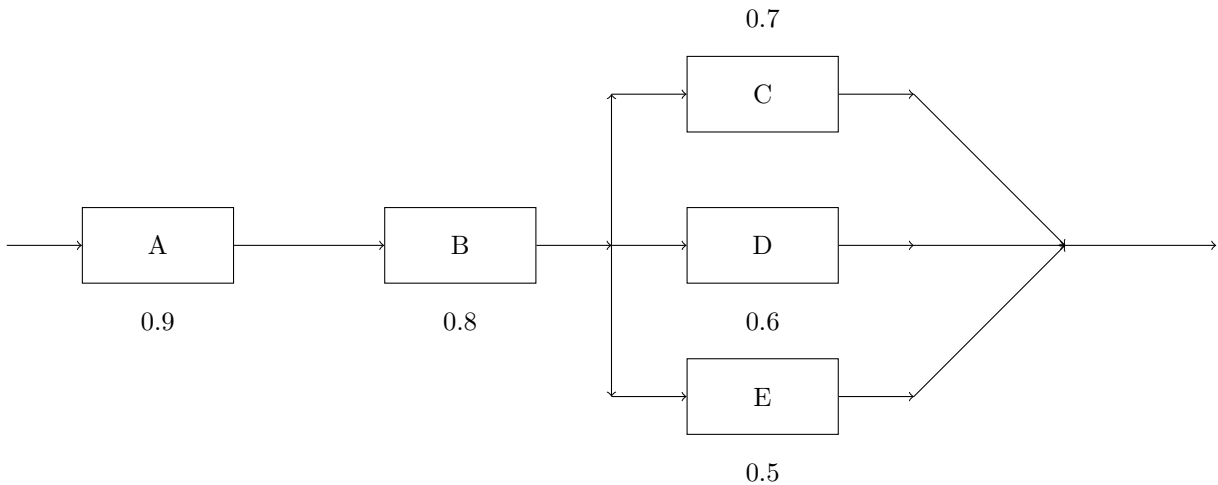


Probability and Statistics (MT-2005)

Homework # 02(d)

Probability & Conditional Probability

1. There is a 1% probability for a hard drive to crash. Therefore, it has two backups, each having a 2% probability to crash, and all three components are independent of each other. The stored information is lost only in an unfortunate situation when all three devices crash. What is the probability that the information is saved?
2. In the system, given by the below figure, if components A, B, C, D, and E function properly with probabilities 0.9, 0.8, 0.7, 0.6, and 0.5, respectively.



$$R_{sys} = (R_A \times R_B) \times [1 - (1 - R_C)(1 - R_D)(1 - R_E)]$$

- (a) Compute the system reliability
 - (b) What will be the impact of failure of component D.
 - (c) If the system needs a minimum reliability of 0.5, what is the minimum reliability value R_C must have to meet this requirement (keeping all other values the same)?
 - (d) If the reliability of Component E is improved from 0.5 to 0.8, calculate the new system reliability and compare it to the original value.
 - (e) Which single component's reliability improvement would result in the greatest increase in overall system reliability? Justify your answer with calculations.
 - (f) If a redundant backup is added in parallel to Component D with the same reliability 0.6 recalculate the new system reliability.
3. Two out of six computers in a lab have problems with hard drives. If three computers are selected at random for inspection, what is the probability that none of them has hard drive problems?

4. A computer maker receives parts from three suppliers, S_1 , S_2 , and S_3 . Fifty percent come from S_1 , twenty percent from S_2 , and thirty percent from S_3 . Among all the parts supplied by S_1 , 5% are defective. For S_2 and S_3 , the portion of defective parts is 3% and 6%, respectively.
- (a) What portion of all the parts is defective?
 - (b) A customer complains that a certain part in her recently purchased computer is defective. What is the probability that it was supplied by S_1 ?
5. A computer assembling company receives 24% of parts from supplier X, 36% of parts from supplier Y, and the remaining 40% of parts from supplier Z. Five percent of parts supplied by X, ten percent of parts supplied by Y, and six percent of parts supplied by Z are defective. If an assembled computer has a defective part in it, what is the probability that this part was received from supplier Z?
6. A problem on a multiple-choice quiz is answered correctly with probability 0.9 if a student is prepared. An unprepared student guesses between 4 possible answers, so the probability of choosing the right answer is $1/4$. Seventy-five percent of students prepare for the quiz. If Mr. X gives a correct answer to this problem, what is the chance that he did not prepare for the quiz?